Real-Time Swahili Speech Analysis System

Bridging Communication Gap in Africa

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Bridging Voices, Connecting Communities

Introduction

Background:
Swahili, a widely spoken language in East Africa, has limited resources for advanced speech processing. This gap hinders the development of automated tools for speech-to-text conversion, sentiment analysis, and summarization.

Problem Statement:
The lack of real-time Swahili speech analysis tools makes it difficult to extract meaningful insights from audio data, affecting fields such as education, business intelligence, and media monitoring.

Motivation:
There is a growing need for analyzing Swahili audio data for customer feedback analysis, media monitoring, and educational support. This research aims to bridge the gap by developing a system capable of real-time speech processing.

Objectives:
•Convert spoken Swahili into text using speech-to-text (STT) technology.
•Analyze sentiment from transcribed speech.
•Summarize transcriptions to extract key insights.

Methods

Data Source:
The Mozilla Common Voice Swahili dataset (Version 11.0) was used, containing diverse speech samples from native Swahili speakers.

Data Collection & Preprocessing:
•Noise reduction and silence trimming Feature extraction using Mel-Frequency Cepstral Coefficients (MFCCs) and spectral centroid. Text normalization, including punctuation removal and stopword elimination.

Model Selection:
•**Speech-to-Text:** Whisper model by OpenAI, known for high transcription accuracy.
•**Sentiment Analysis:** DistilBERT fine-tuned on SST-2 for sentiment classification.
•**Text Summarization:** T5 model used for generating concise summaries
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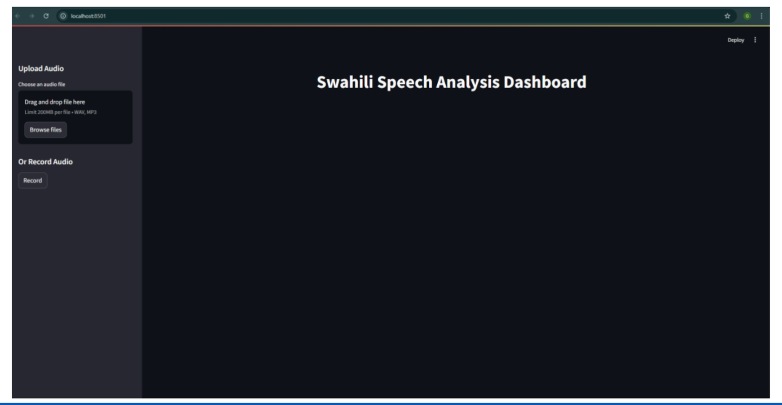
System Architecture:
1.Audio Input.
2.Preprocessing. Speech-to-Text..
3.Sentiment Analysis. Text Summarization.

Deployment:
A web interface was built using Streamlit for real-time interaction.



Data Analysis

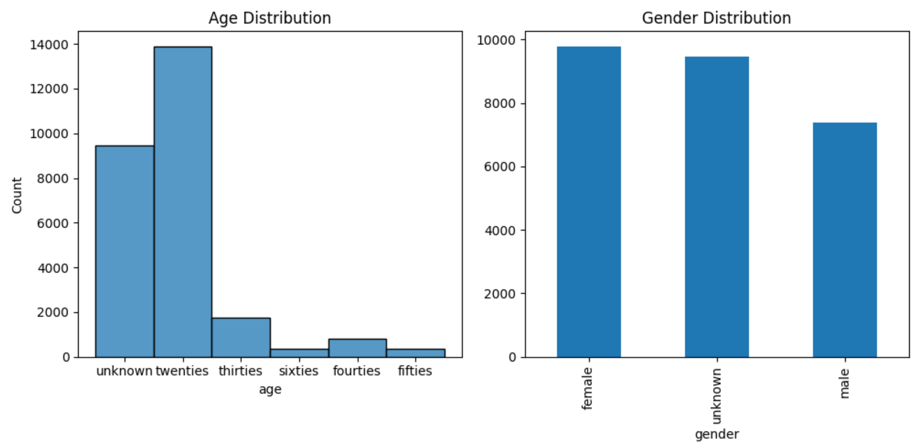


Fig1:Demographic Dynamics of the Data

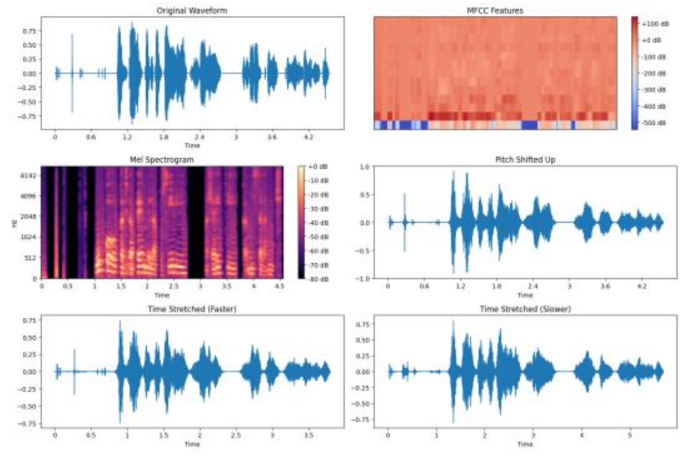


Fig2:Feature Engineering

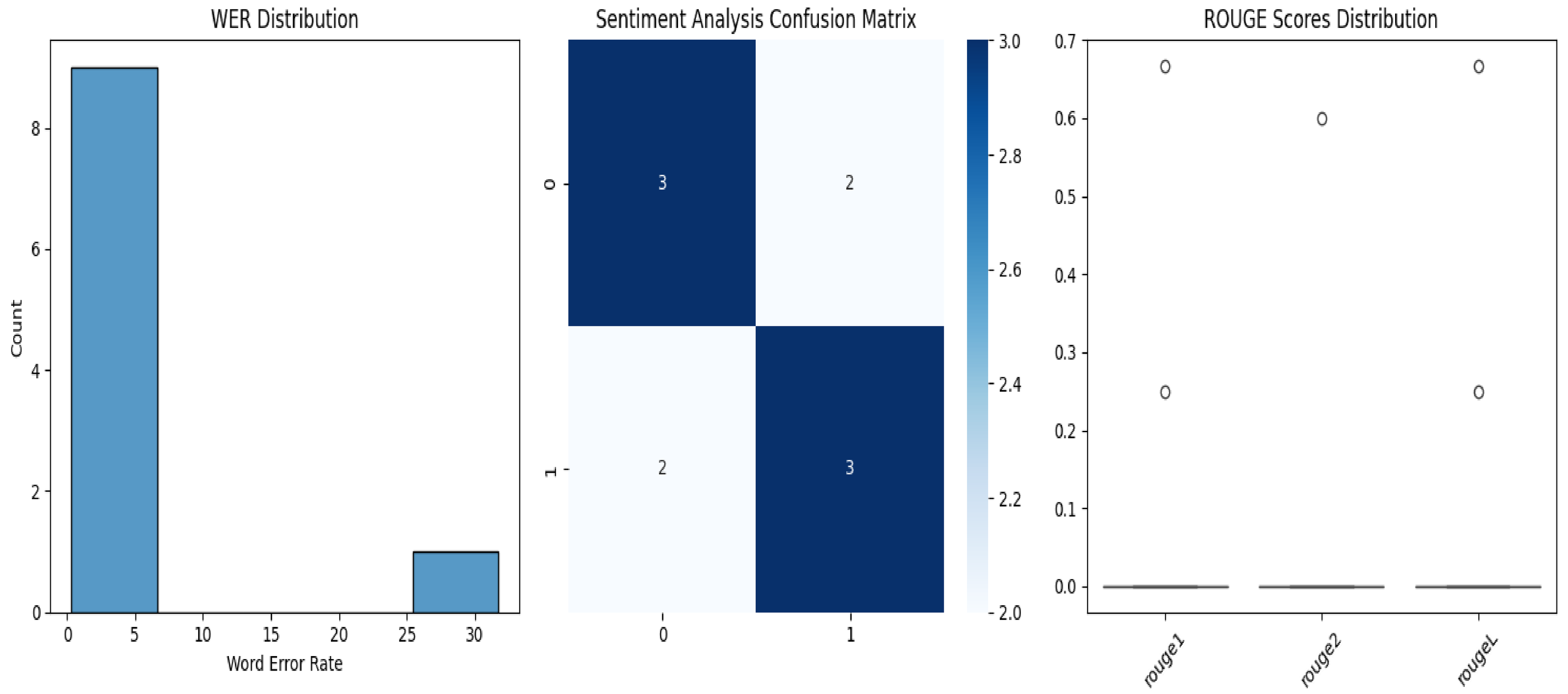


Fig3:Model Evaluation

Model	Evaluation Metric
Speech Recognition	Average WER: 3.9948
Sentiment Analysis	Accuracy: 0.6000
	F1 Score: 0.6000
Text Summarization (ROUGE Scores)	ROUGE-1: 0.0917
	ROUGE-2: 0.0600
	ROUGE-L: 0.0917

Interpretation:
The model performs well but can be improved by fine-tuning on Swahili-specific data.

Results

Findings:
•The system successfully transcribed Swahili speech with high accuracy.
•Sentiment classification identified emotions effectively.
•Summarized text retained key information concisely.

Qualitative Analysis:
•Users found the system efficient for real-time applications.
•Generated summaries improved readability and accessibility.
•Sentiment classification aligned well with expected emotional tones.

Conclusion

This research presents a pioneering approach to real-time Swahili speech processing, demonstrating the potential of AI in underrepresented languages.

Practical Applications:
•Enables multilingual AI solutions for speech processing.
•Improves accessibility in technology for Swahili speakers.
•Supports further advancements in speech-driven analytics.

References

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• Lin, C. Y. ROUGE: A Package for Automatic Evaluation of Summaries. ACL, 2004.